

# CS 4725: Research Seminar on Scalable Learning Systems

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**Kubilay Atasu** & Lydia Y. Chen

# Scalable Learning Systems @ TU Delft



**Software Technology Dept.**  
**Data Intensive Systems Sect.**



Before TU Delft!



**Kubilay Atasu**



Lydia. Y. Chen

# Teaching Team

- Dr. Kubilay Atasu, Associate Professor
  - Lectures, Projects, Presentations, Homeworks & Coordination
- Dr. Rui Wang, Postdoctoral Researcher
  - Projects & Guest Lecture
- Mr. Dennis Heijmans, MSc. Thesis student
  - Homeworks & Presentations

# Ongoing MSc theses in my group

- ▷ **Rodrigo Alvarez Lucendo**, Scaling Laws for Memorization in LLMs
- ▷ **Selin Ceydeli**, Privacy-Preserving Subgraph Federated Learning Systems
- ▷ **Dennis Heijmans**, Adversarial Robustness of Multigraph Neural Networks
- ▷ **Zeryab Alam**, Time-Series Foundation Models for Geothermal Energy Systems
- ▷ **Andrea Li**, Subgraph Heterogeneity in Federated Graph Neural Networks
- ▷ **Danae Savvidi**, Temporal Graph Neural Networks for Financial Crime Analysis

# Your Background

# Expected Background

- Basic deep machine learning models → Knowledge
- System/code deployment → Engineering Experience
- [Dive into Deep Learning](#), Chapters 1 – 6,  
A. Zhang, Z. J. Lipton, M. Li, and A. J. Smola.  
We cover topics from Chapters 11 – 13

# Class Organization

# Overview - Lecture Schedule

**5 weeks of lectures + student presentations**

## Topics

- Introduction to Distributed Machine Learning Systems
- Acceleration of Machine Learning and Deep Learning
- Acceleration of Large Language Models
- Federated Learning Systems
- Attacks and Defenses in ML Systems
- Paper Presentations
- Final Project Presentations

# Overview – Assessment

- **Paper presentations (20%):** Each group will present a paper and lead the discussion (weeks 7-8)
  - 10-12 mins to present the paper and 5-6 mins for the Q&A
- **Research project (60%):** Each group will reproduce the work done in a recent article from a top-tier conference and try to improve its results
  - Deliverables: A final project report and a final project presentation in week 9
  - A draft of the project report in week 8 (not graded)
- **Individual homeworks (20%):** written (due in weeks 4 and 5)

## Paper Presentation Rubric

| Item                                                                             | Percentage (%) |
|----------------------------------------------------------------------------------|----------------|
| Motivation/ Problem statement (why this is an important and challenging problem) | 20             |
| Proposed solutions/ contribution (The details and novelty characteristics )      | 40             |
| Evaluation (How is the proposed solution evaluated )                             | 25             |
| Leading the in-class discussion                                                  | 15             |

# Project (I)

- **Objective**

- Reproduce one of the key aspects of the paper selected
- Improve/validate the algorithm by your proposed solution

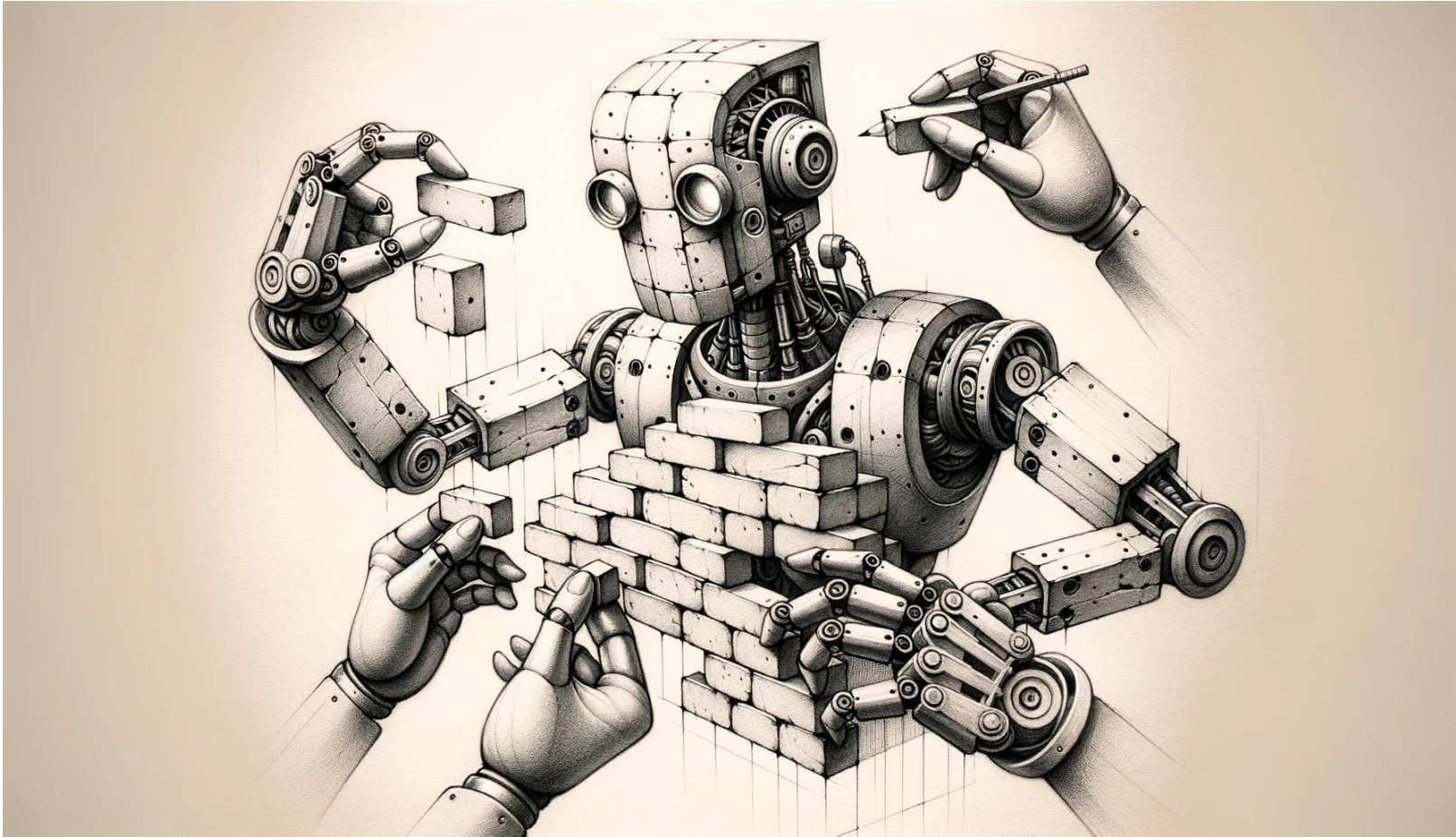
# Project (II)

## ○ Milestones

- **Project proposal** (mandatory, *ungraded*): Due in week 4
- **Midterm review meeting:** (mandatory, *ungraded*): In week 6.
- **Draft of the project report:** (mandatory, *ungraded*): Due in week 8.
- **Final project report** (mandatory, *graded*): Due in week 9.
- **Final project presentation** (mandatory, *graded*): Due in week 9.

# Todo

- **Form groups of three students**
  - Same Groups for Projects and Presentations
- **See the paper list on BrightSpace**
  - Separate lists for Projects and Presentations
  - Select your papers by Week 3
- **Request access to DelftBlue or DAIC**
  - The TAs will provide the instructions



# Systems for Foundation Models, and Foundation Models for Systems.

Chris Ré Stanford, NeurIPS 2023 Keynote Talk

# Some Questions in Systems for AI and AI for Systems

▷ How do FMs change the **systems** we build?

*FMs help “death by 1000 cuts” problems.*



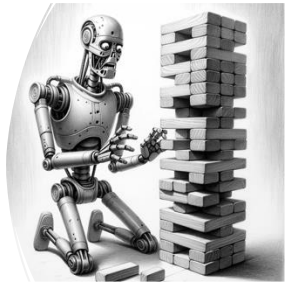
▷ How efficient can we make **current** models?

*Classical database systems ideas for AI: Flash Attention*



▷ Are there **new** models that are more efficient?

*Some exciting new models like S4.*



# Autoregressive Language Models

**Simple, Old idea.** Complete the sentence “*The mouse ate the*” ranked by probability learned from the corpus.

$$p(\text{“the”, “mouse”, “ate”, “the”, “cheese”}) = 0.10$$

$$p(\text{“the”, “mouse”, “ate”, “the”, “mouse”}) = 0.01$$

$$p(\text{“the”, “mouse”, “ate”, “the”, “house”}) = 0.0001$$



Tatsu  
Hashimoto



Percy  
Liang

## Neural Language Models.

- ▷ Neural nets “compactly” represent that probability function
- ▷ Generate. Generate answers using conditional probability.
- ▷ Train. Every single token in a sentence is an example.

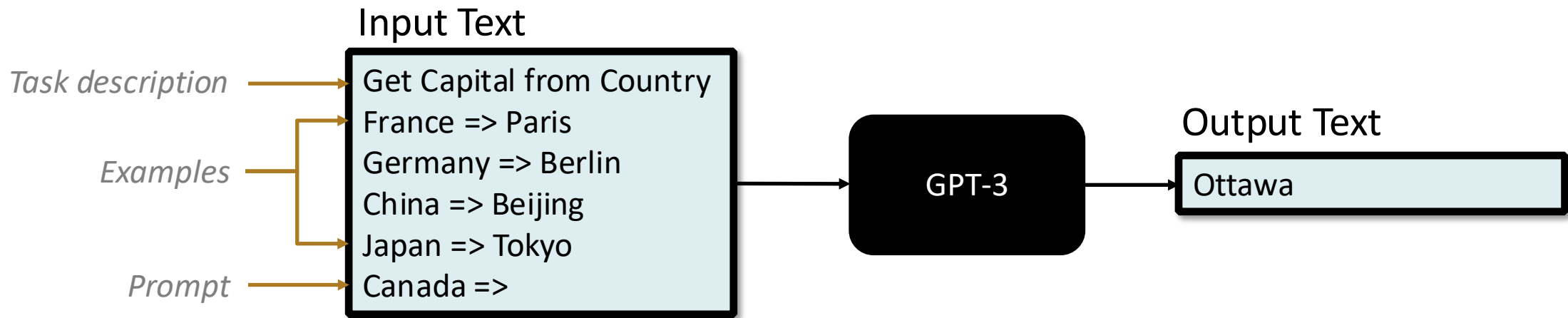
# Foundation Models



## Foundation Models

- ▷ Autoregressive neural language models
- ▷ **Very large** number of parameters
- ▷ **Very large** training corpuses

Emergent Behaviors: Generalize to new tasks with NO finetuning (Few-shot)



Natural language text in and out. Awesome!

# Emergent Behaviors

It works on many different language tasks

- Translation

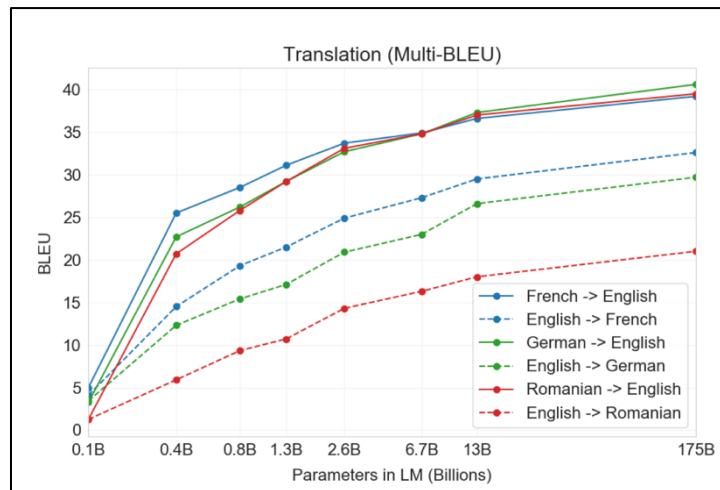
Input Text

Translate English to French  
Cheese => fromage  
Wine =>

GPT-3

Output Text

Vin



# Emergent Behaviors

It works on many different language tasks

- Translation
- Trivia / Question Answering

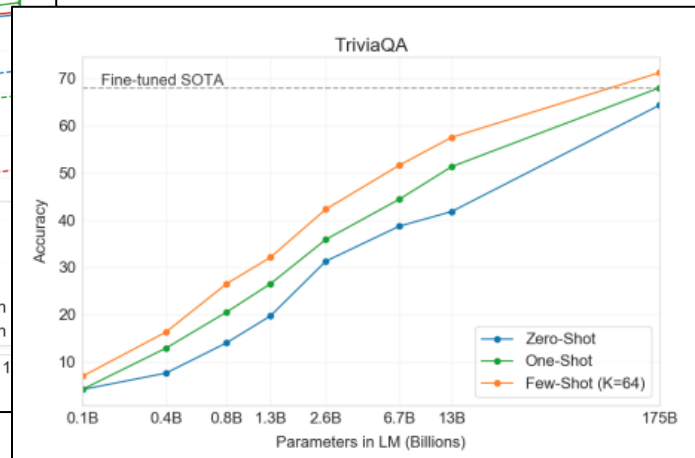
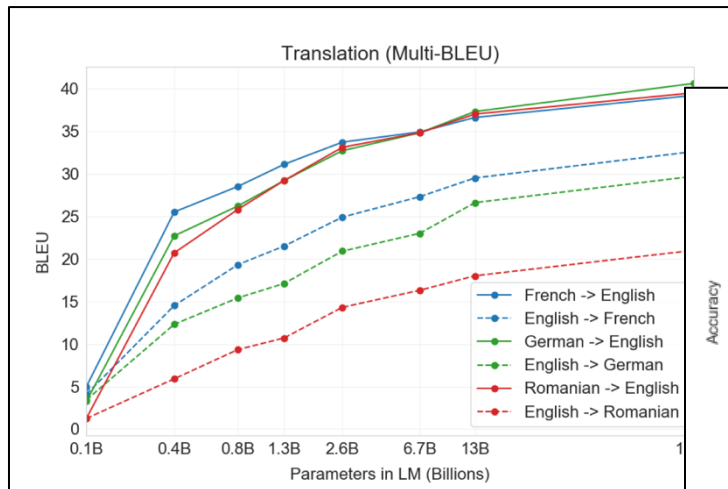
Input Text

Q: 'Nude Descending A Staircase' is perhaps the most famous painting by which 20th century artist?  
A:

GPT-3

Output Text

Marcel Duchamp



# Emergent Behaviors

It works on many different language tasks

- Translation
- Trivia / Question Answering
- Arithmetic
- And many more.....

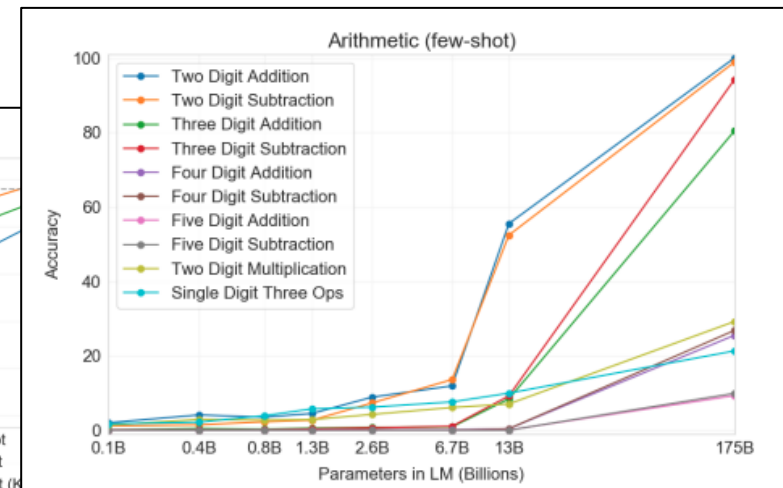
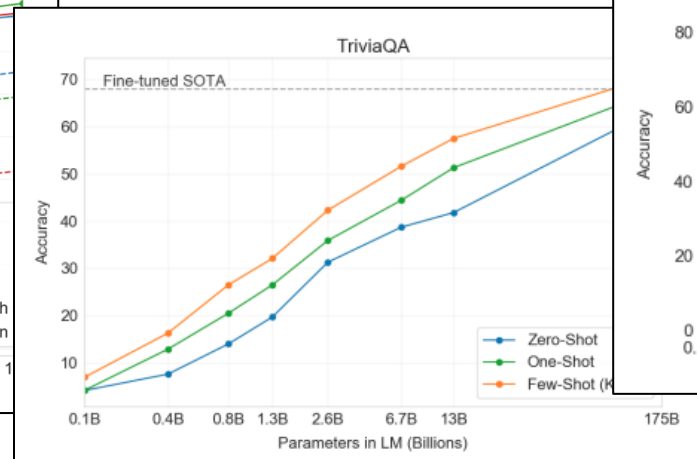
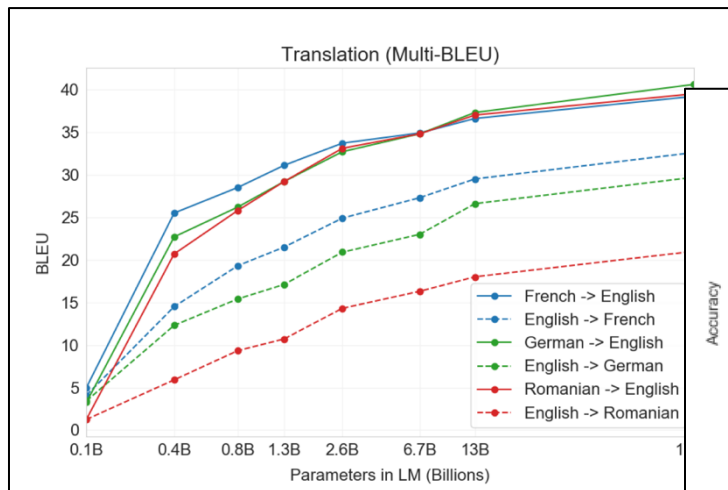
Input Text

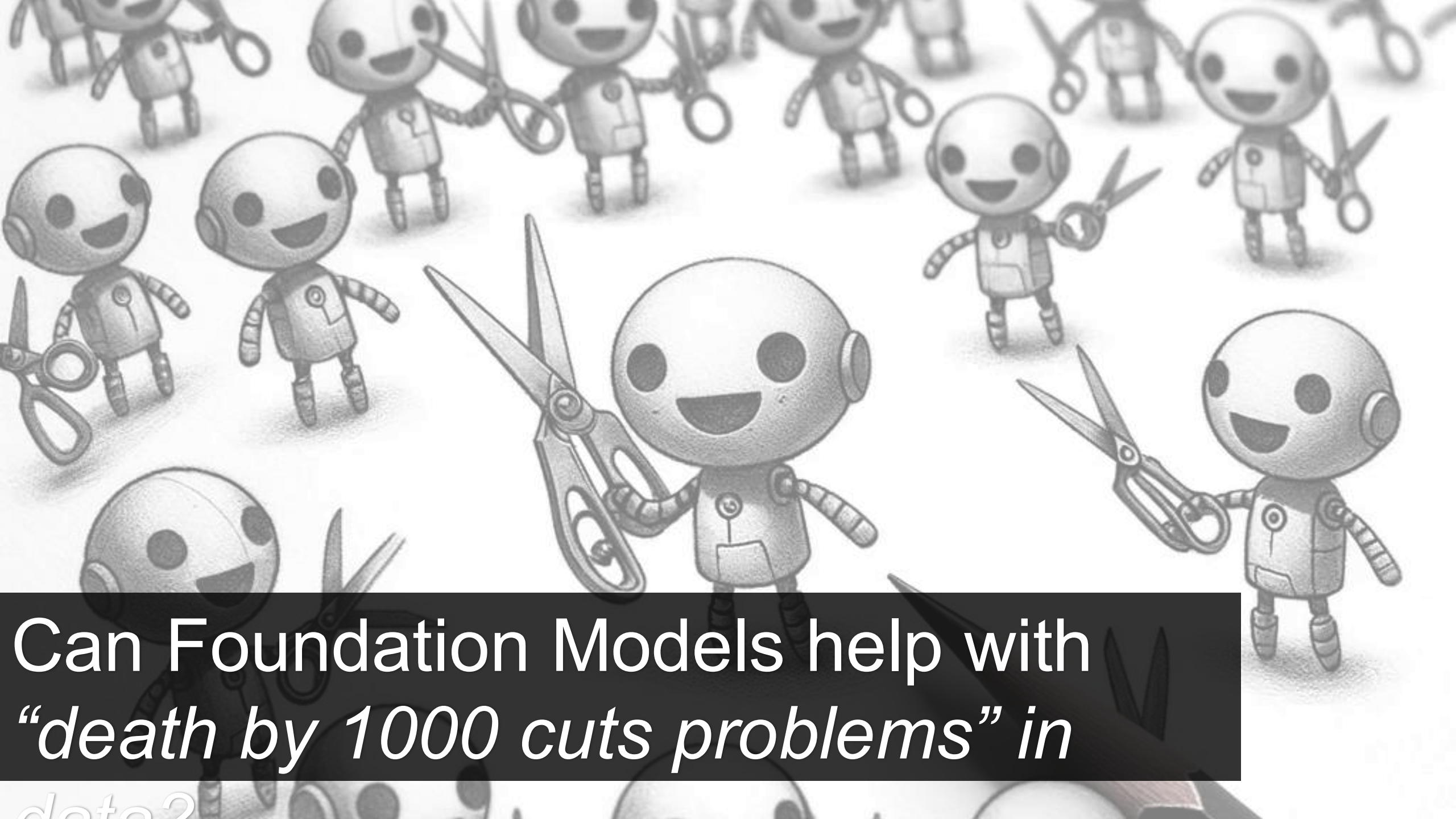
Q: What is 17 minus 14?  
A:

GPT-3

Output Text

3





Can Foundation Models help with  
“death by 1000 cuts problems” in

data?



# Example: Data Cleaning

▷ Goal: Detect and repair errors in structured data

▷ Diverse errors:

- Typos and formatting
- Conflicting values
- Outlier values

|    | DBAName           | AKAName   | Address          | City           | State | Zip          |
|----|-------------------|-----------|------------------|----------------|-------|--------------|
| t1 | John Veliotis Sr. | Johnnyo's | 3465 S Morgan ST | <b>Chicago</b> | IL    | <b>60608</b> |
| t2 | John Veliotis Sr. | Johnnyo's | 3465 S Morgan ST | Chicago        | IL    | <b>60609</b> |
| t3 | John Veliotis Sr. | Johnnyo's | 3465 S Morgan ST | Chicago        | IL    | <b>60609</b> |
| t4 | <b>Johnnyo's</b>  | Johnnyo's | 3465 S Morgan ST | <b>Cicago</b>  | IL    | 60608        |

Conflicts

Does not obey  
data distribution

Conflict

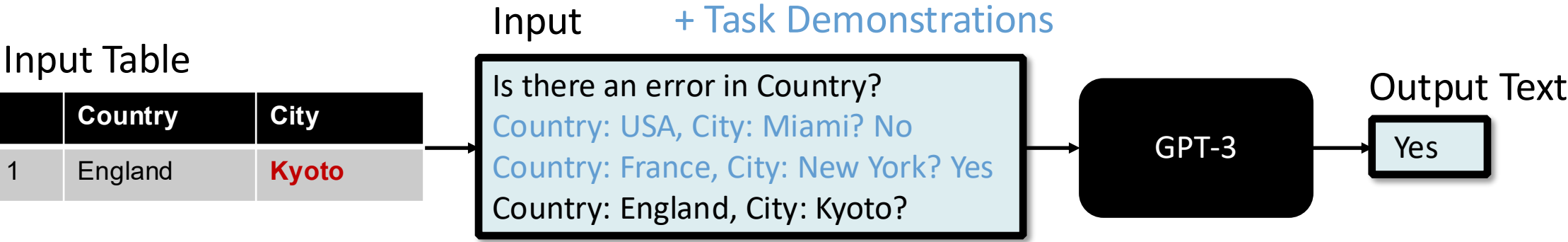
Data cleaning has a “*death by a thousand cuts*” feel:  
*Each problem easy, but sheer variety and breadth hard*



Avanika Narayan



# Foundation Models for Data Tasks



Zero-shot works, but not as good as SotA

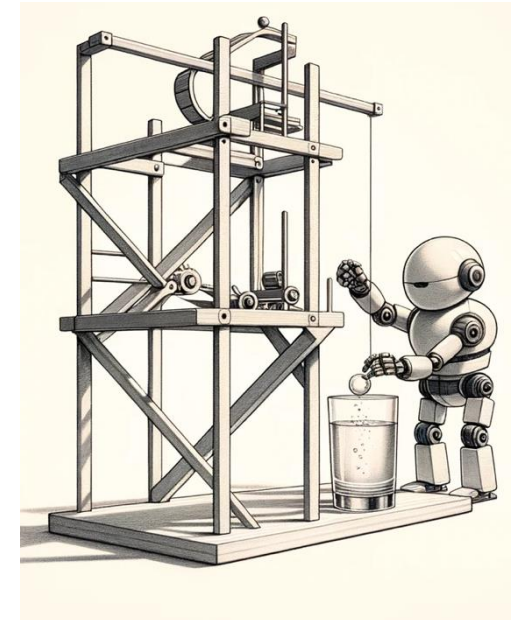
|                   | Entity Matching |      |                | Imputation |      | Error Detection |
|-------------------|-----------------|------|----------------|------------|------|-----------------|
| Dataset           | iTunes-Amazon   | Beer | Walmart-Amazon | Restaurant | Buy  | Hospital        |
| Previous SoTA     | 97.1            | 94.4 | 86.8           | 77.2       | 96.5 | 94.4            |
| GPT-3 (Zero-shot) | 40.3            | 63.6 | 49.1           | 73.3       | 89.2 | 12.1            |
| GPT-3 (Few-shot)  | 98.2            | 100  | 87.0           | 90.0       | 97.1 | 98.1            |

*Few-shot on this model trained only predict words and not retrained! **Wild!***

Aside: an odd, new paradigm?

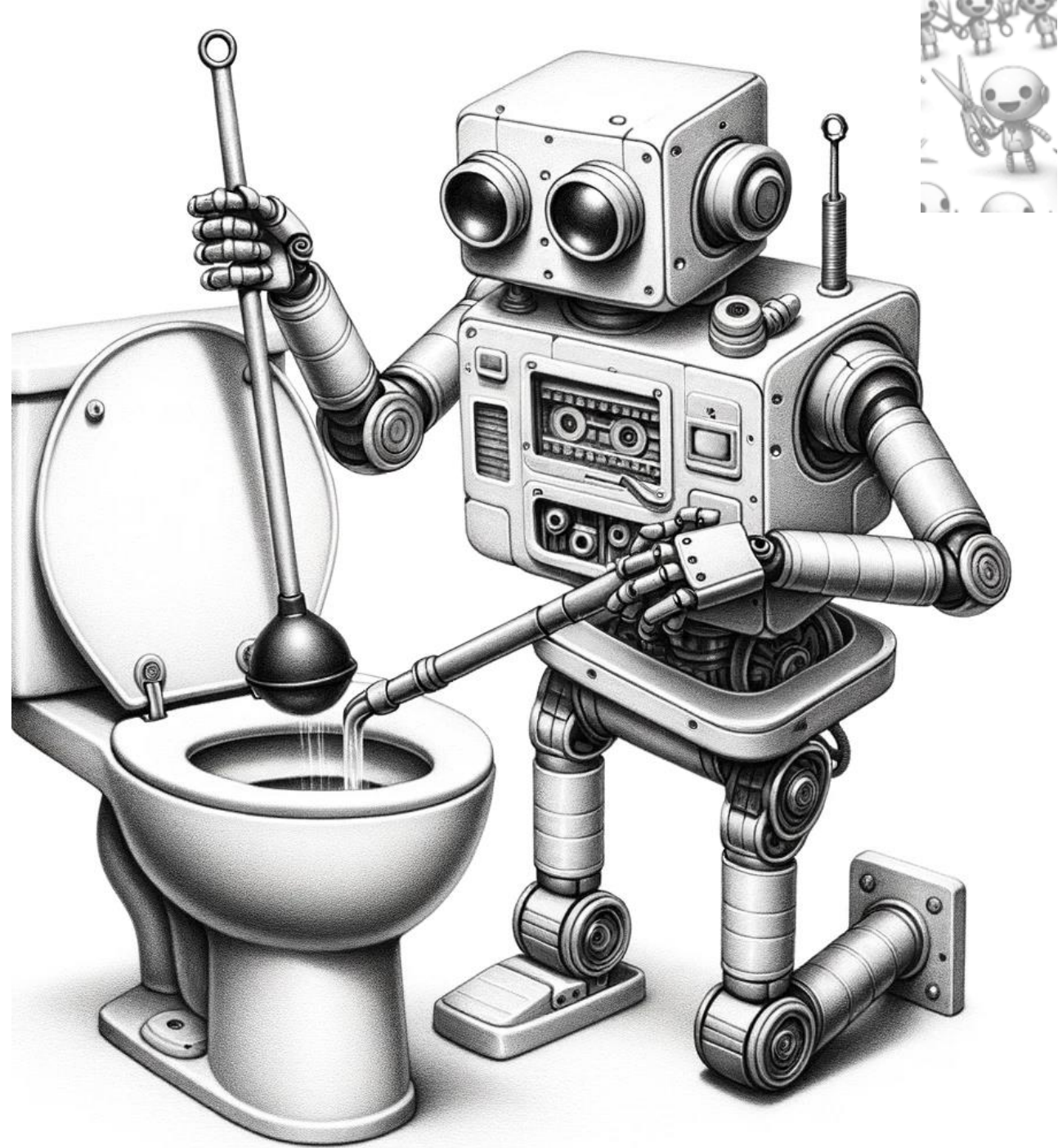
**Classical CS way:** Narrow down to a **specific** problem, then solve it! Precision is the key.

**New FM way:** solve *much* more **general** problem—and use model as a foundation for original problem.



# Summary of Data Plumbing Applications

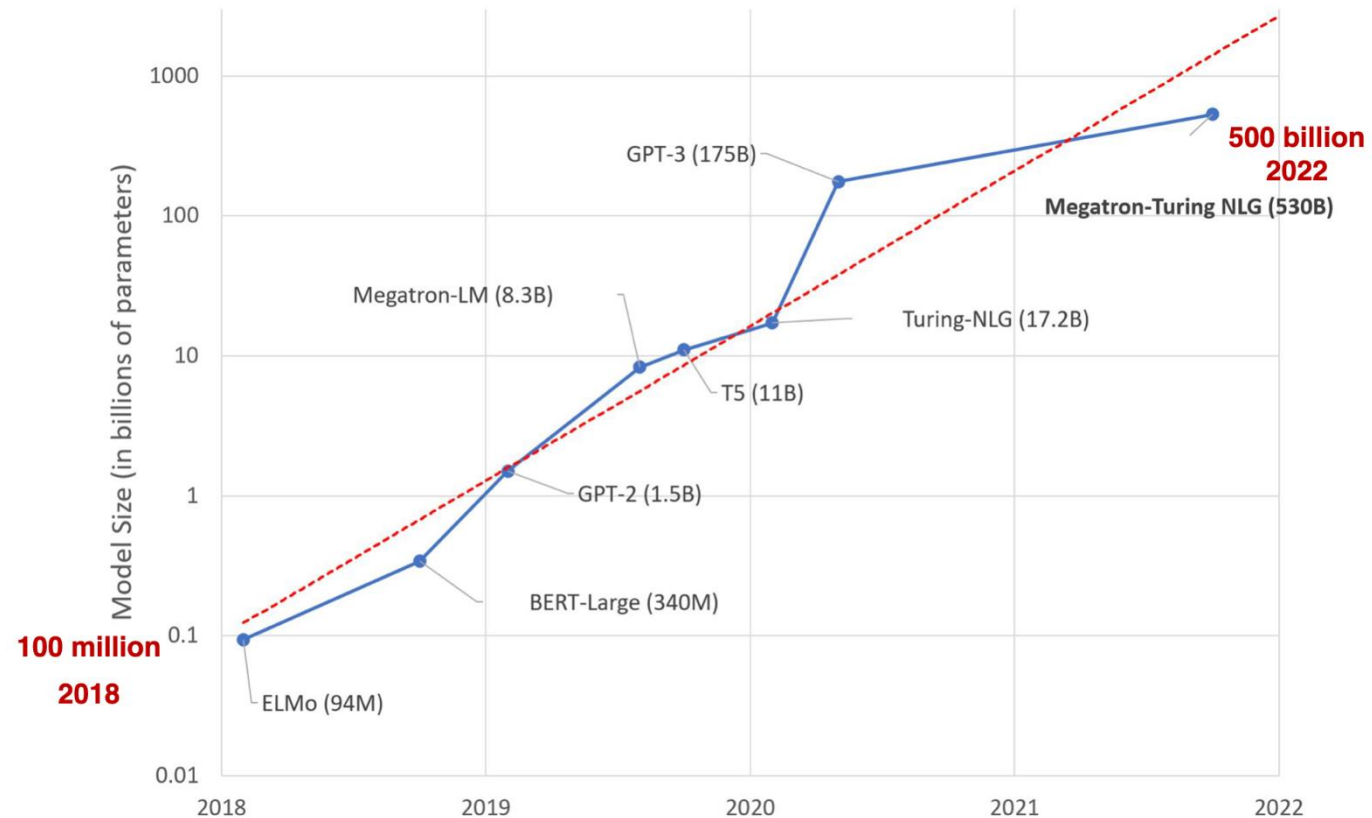
“death by a thousand cuts” problem:  
*1000s of subproblem, each easy but  
breadth of subproblems is hard.*





To build  
new  
applications,  
we may  
need more  
capable  
FMs...

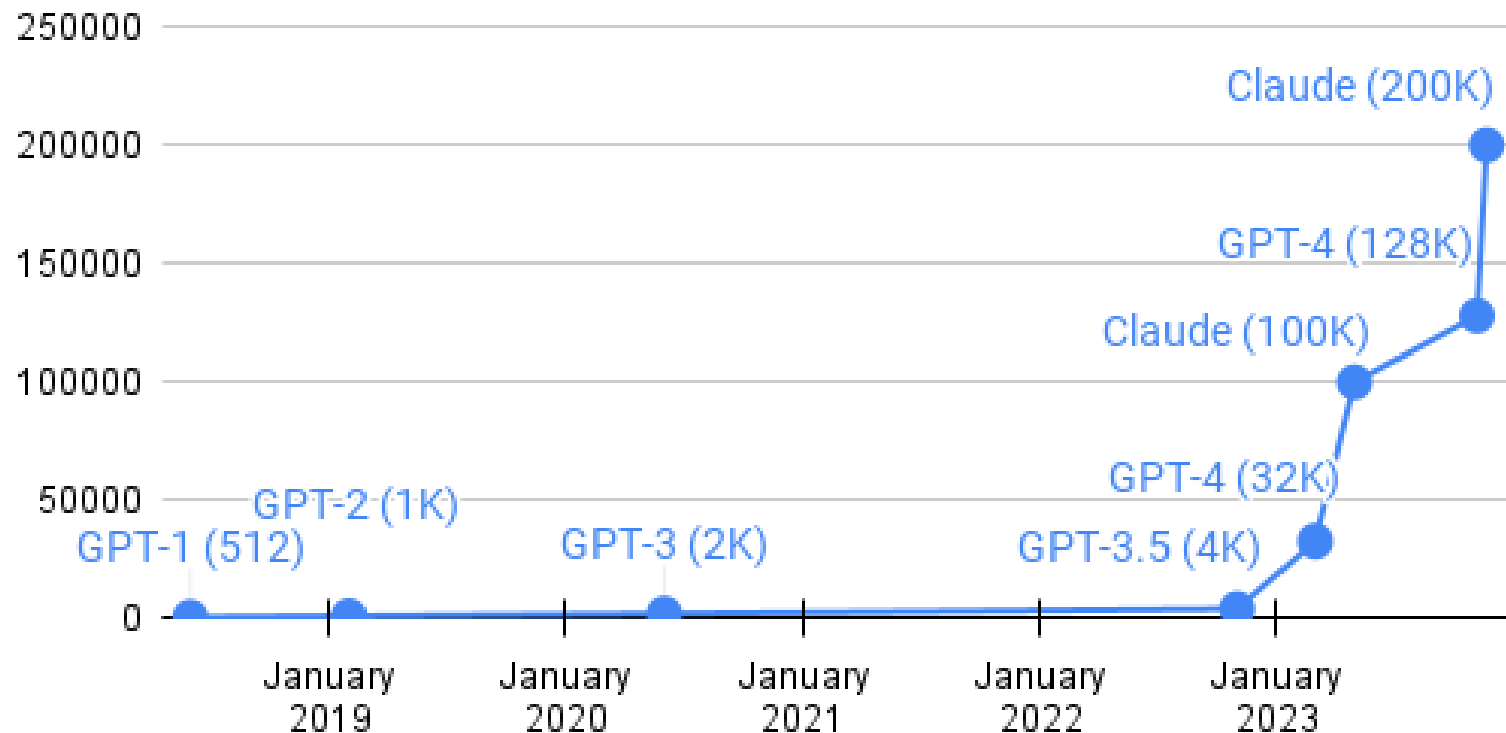
# More capable, often means bigger FMs



Shockingly large growth...

... and more (longer) context in the model!

## Foundation Model Context Length





How efficient can we make current models?

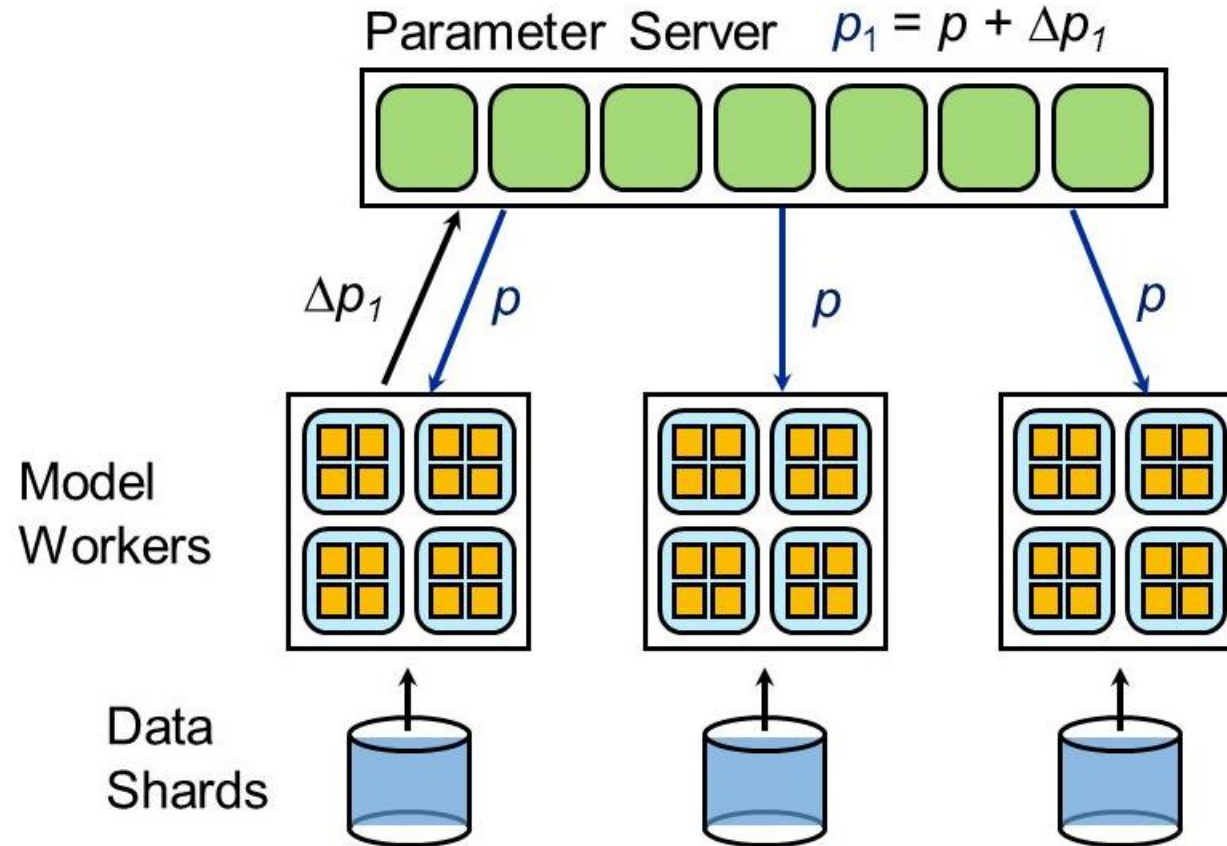
# Goal of This Course

# In This Course

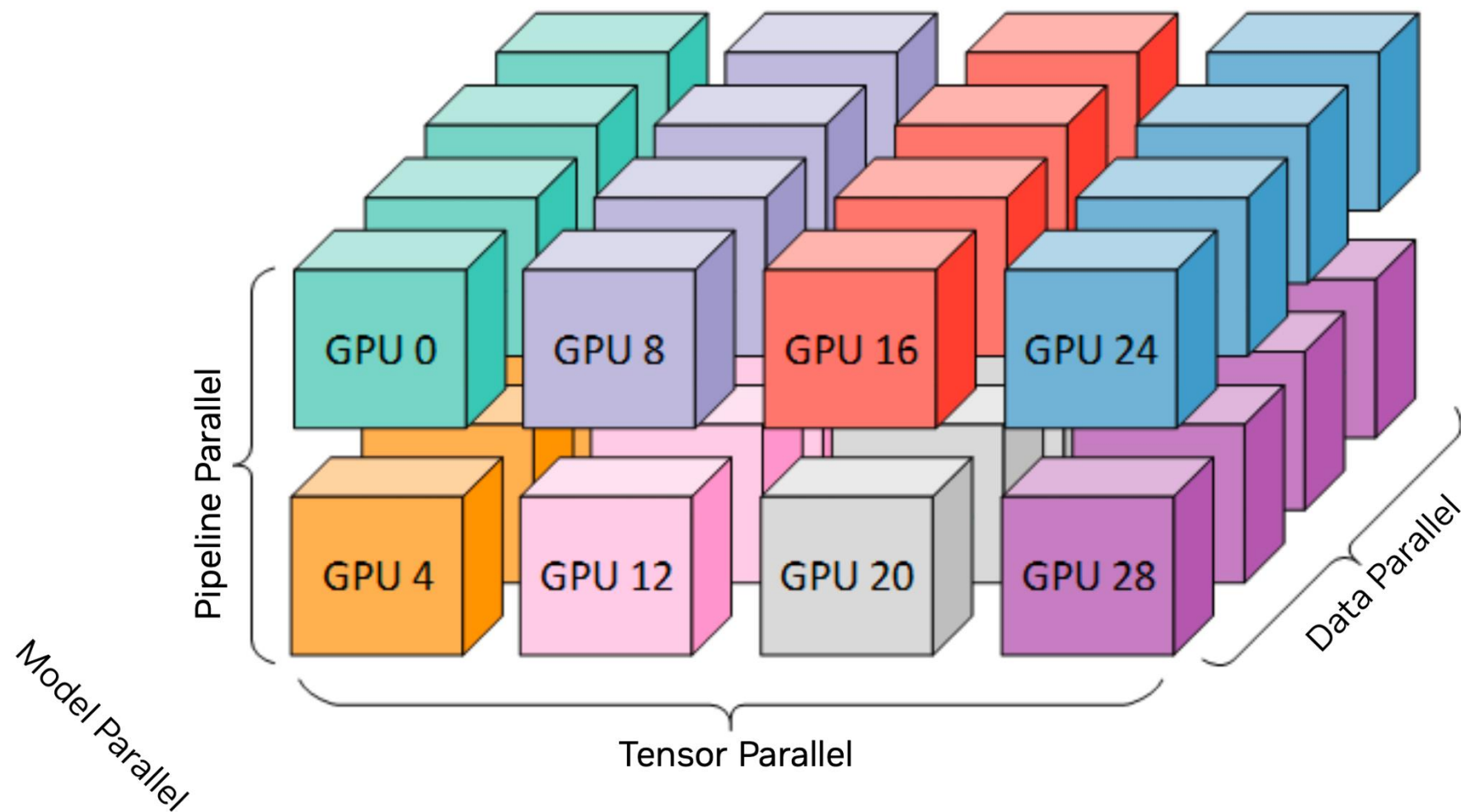
You will learn how to design and build scalable ML systems via paper presentations, discussions, and project prototyping.



# Distributed ML Systems

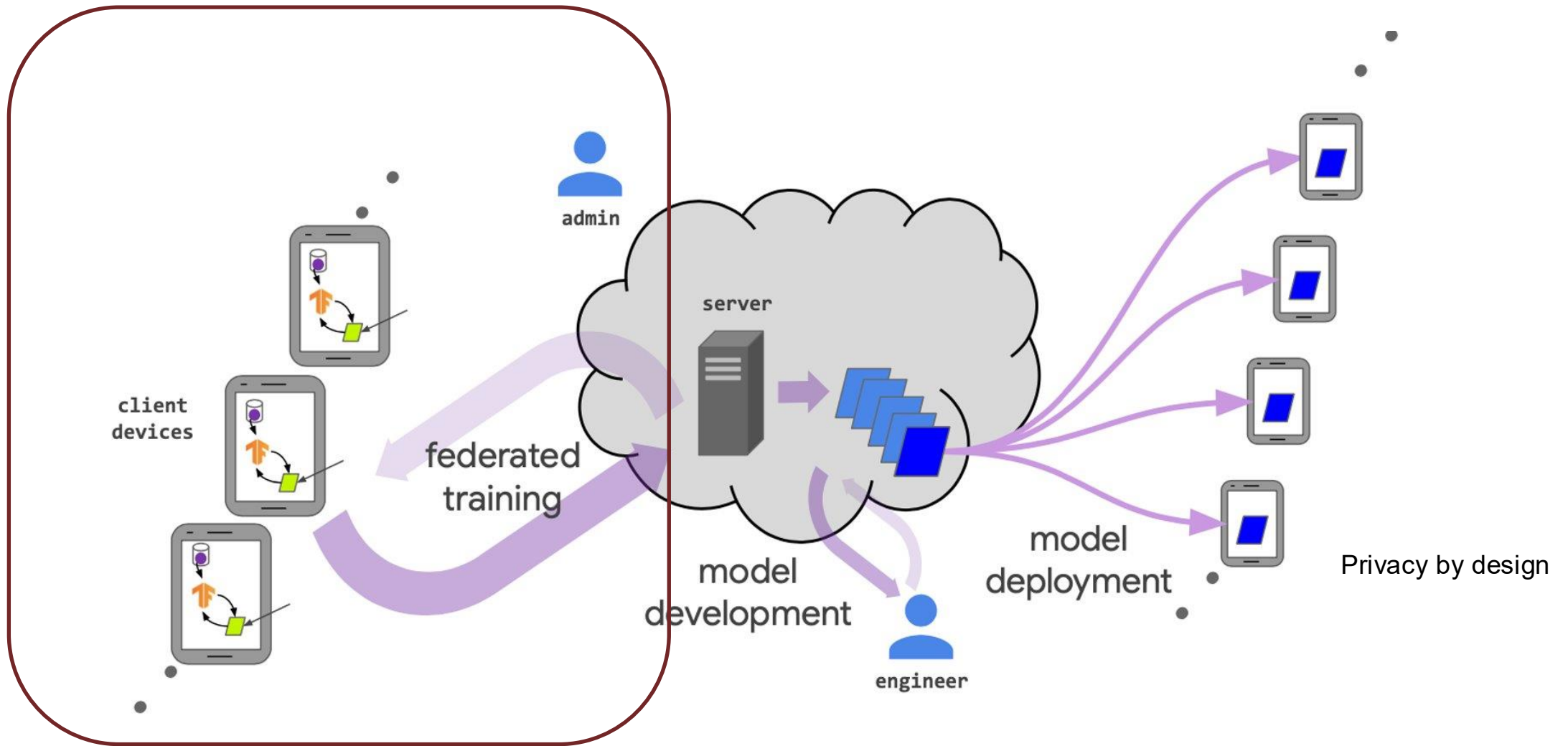


## 3D Parallelism



It takes 20 GPUs to hold one copy of GPT4 model parameters

# Federated ML Systems

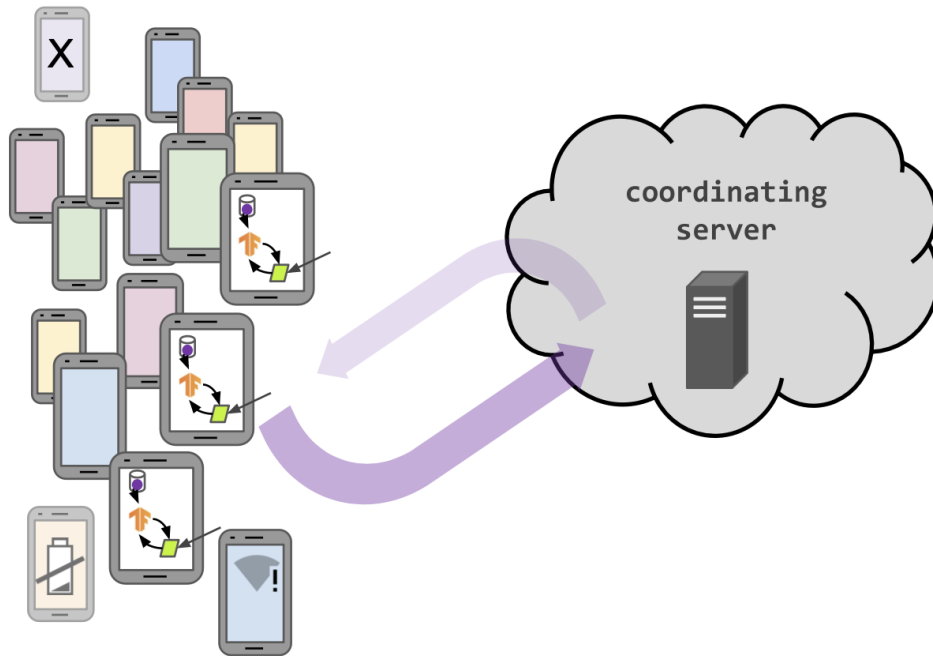


Source: Google federated learning tutorial

# Federated Learning

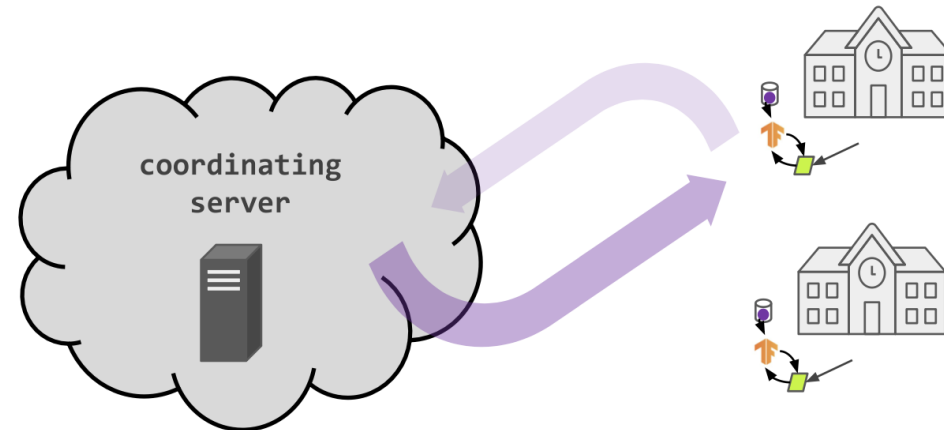
## Cross-device federated learning

millions of intermittently  
available client devices



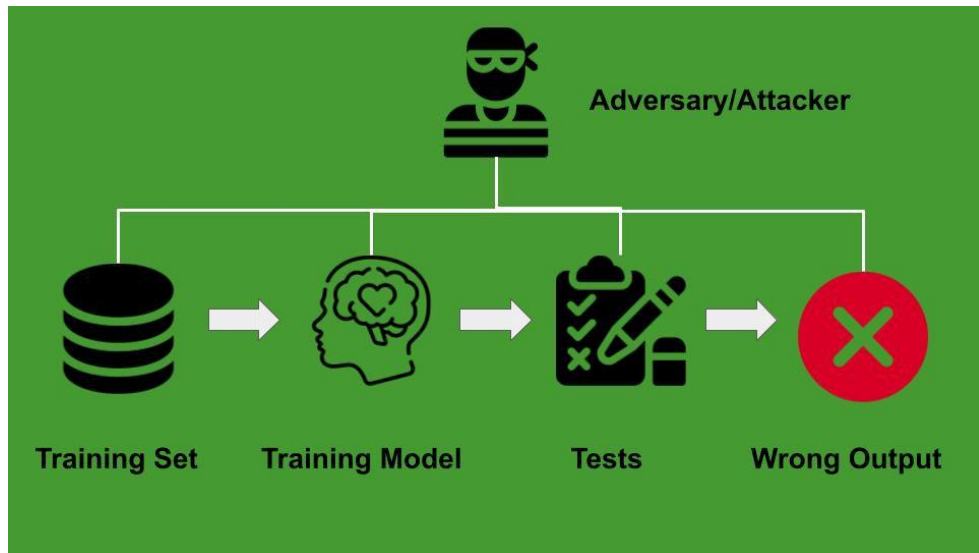
## Cross-silo federated learning

small number of clients  
(institutions, data silos),  
high availability

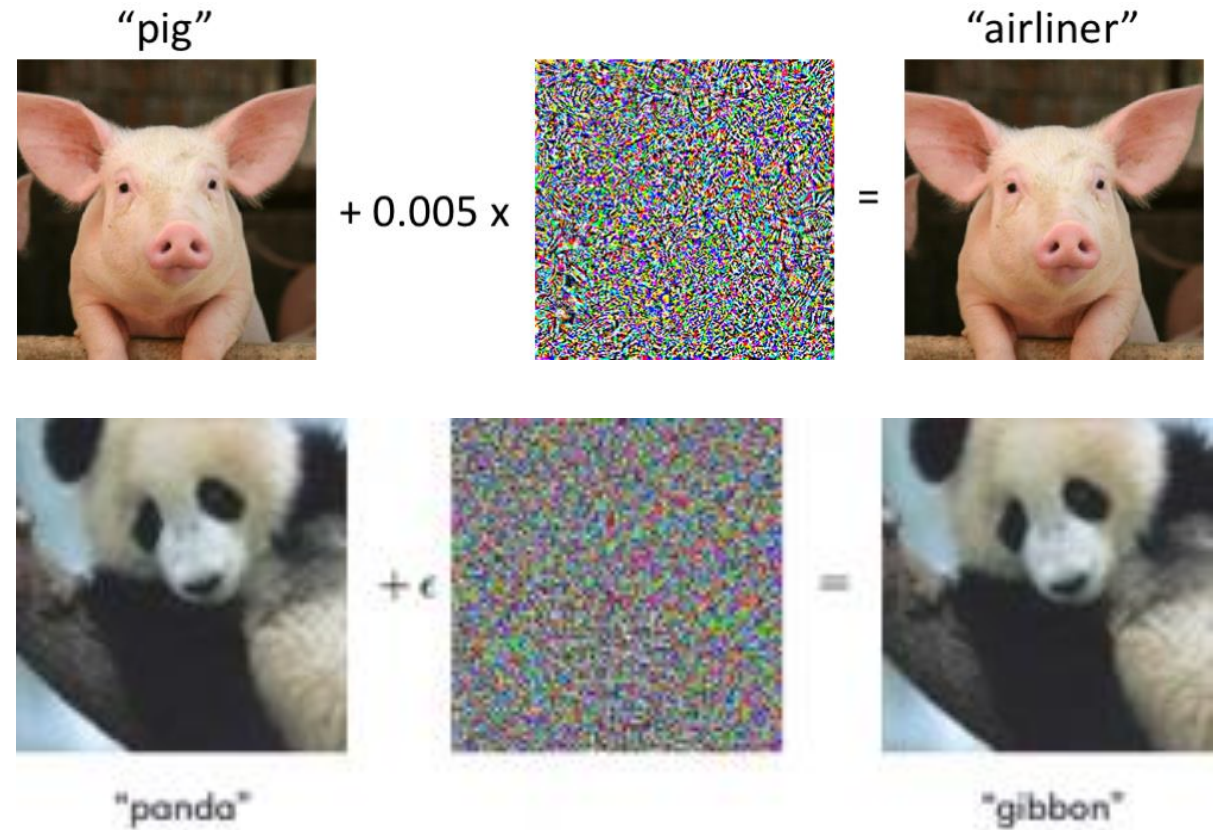


Source: Google federated learning tutorial

# Robustness Against Adversaries



What types of attacks?

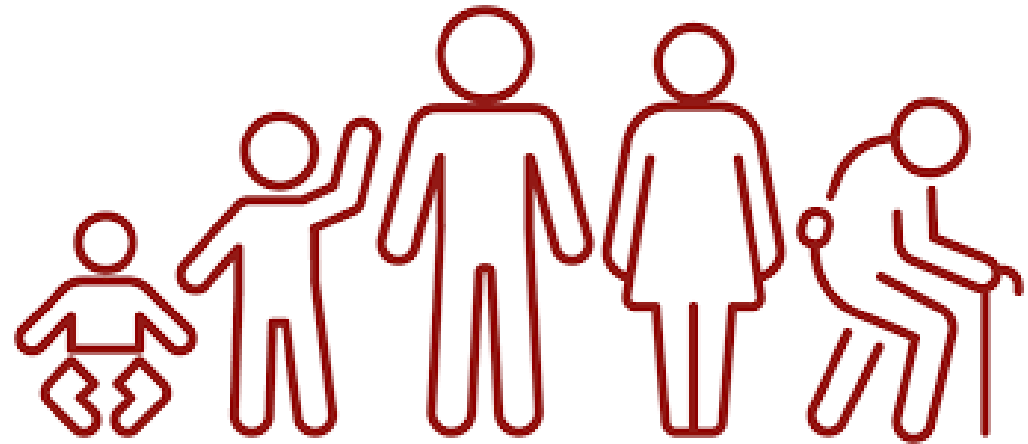


# Federated Generative AI Systems

## Generative models

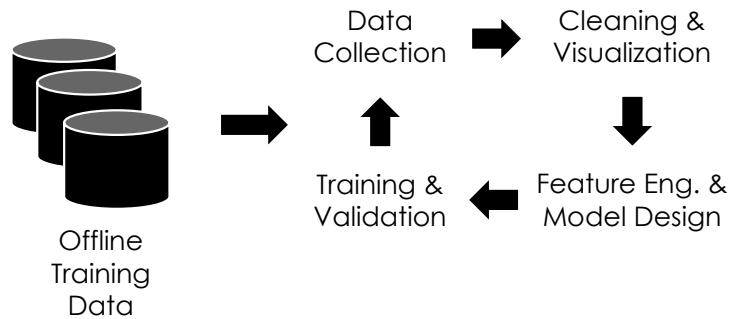


## Personalization



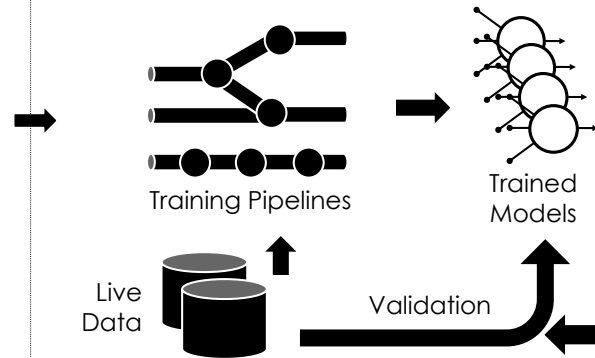
# Traditional ML Lifecycle

## Model Development



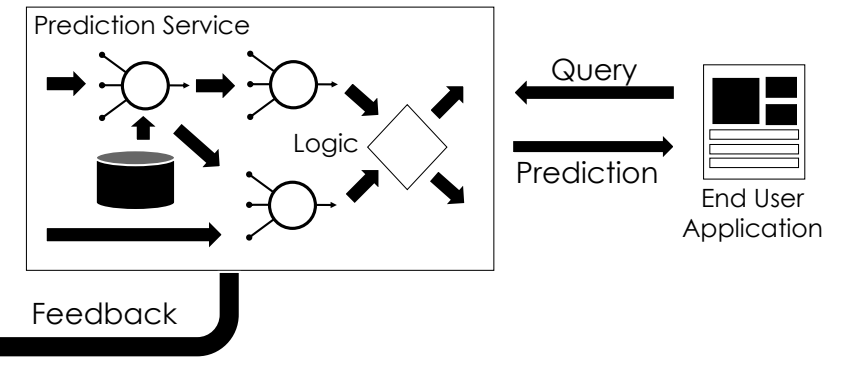
Data Scientist

## Training



Data Engineer

## Inference



Data Engineer

# Pre-training & Fine-Tuning

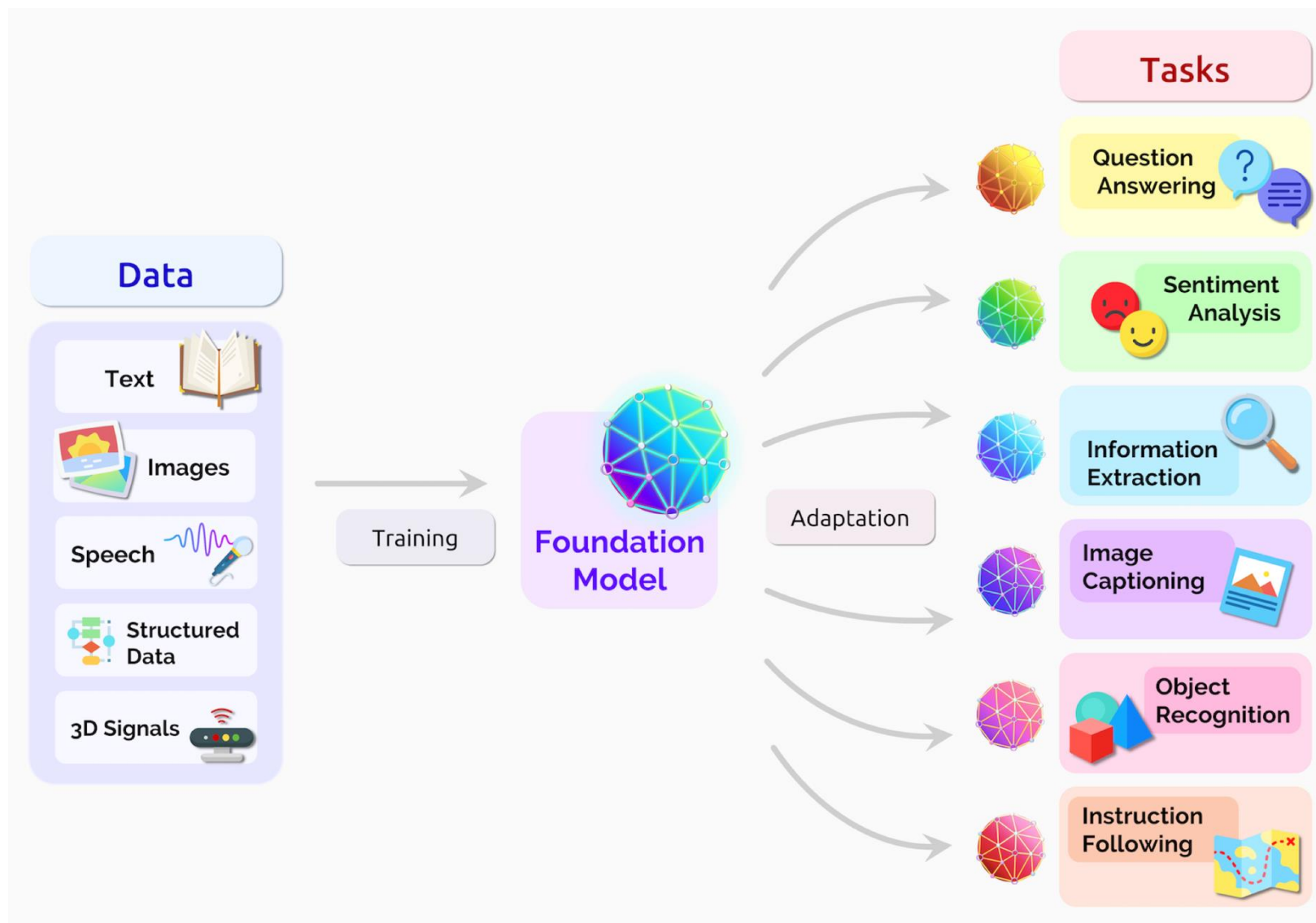
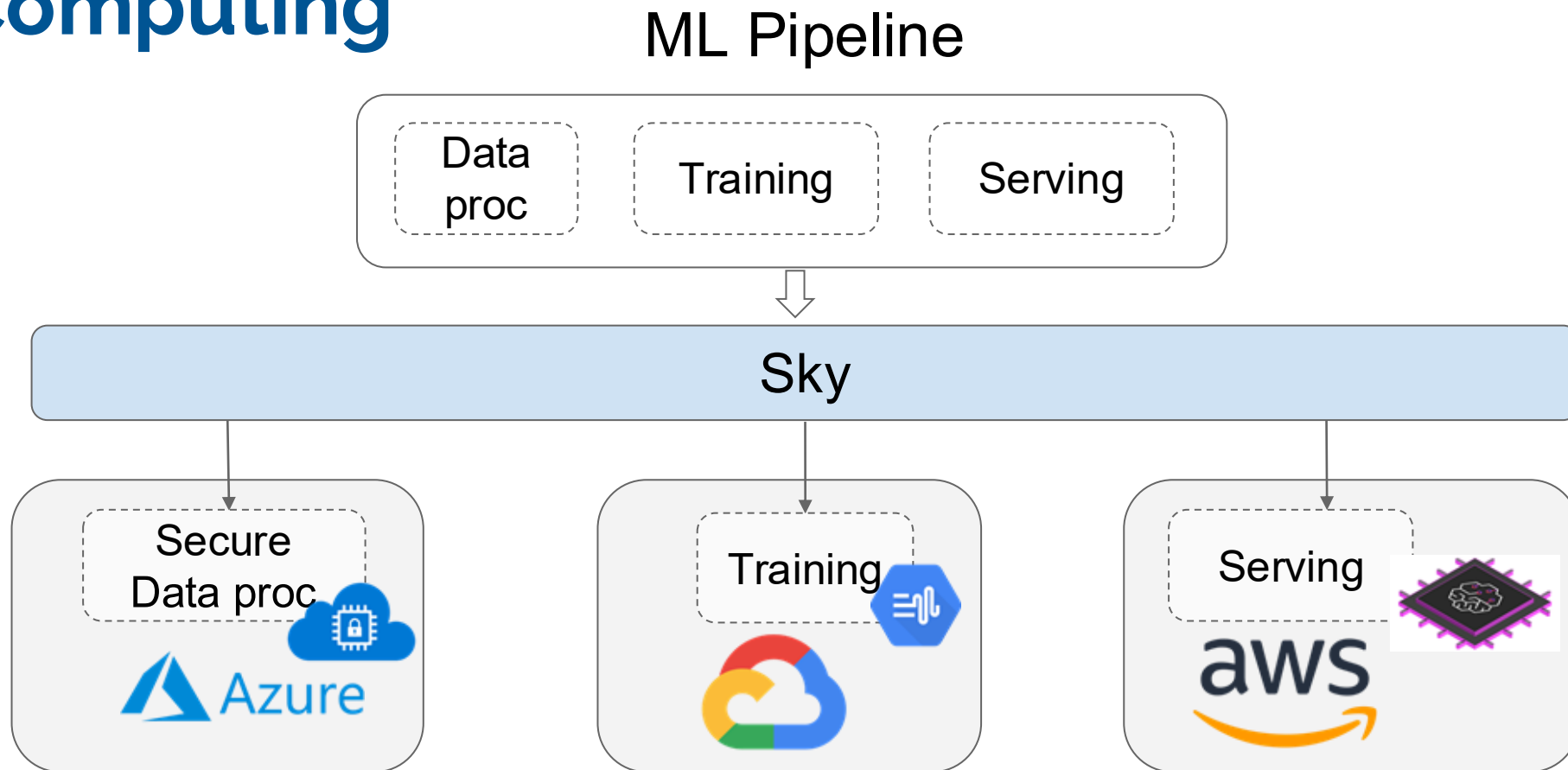


Fig. Source: On the Opportunities and Risks of Foundation Models: <https://arxiv.org/abs/2108.07258>

# Sky Computing



- Use Azure Confidential Computing for secure data processing
- Use Google Cloud for training on TPUs (fast and cheap)
- Use AWS for serving on Inferentia (cheapest)

Thanks!

**Any questions?**

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